

AY: 2025-2026
Course: PPOM 1
Class of 2029
Week 13, Lecture 112

**NEW YORK INSTITUTE
OF TECHNOLOGY**

College of Osteopathic
Medicine

Orthopedic Concerns of the Spine

Presented by:

Matthew Heller, D.O., FAOASM

Associate Professor, Family and Sports Medicine

NYIT College of Osteopathic Medicine

Team Physician, NYIT Athletics

Ringside Physician

Email: mheller@nyit.edu



[@matthewbheller](https://www.instagram.com/matthewbheller)



[@matthewbheller](https://www.x.com/matthewbheller)

Session Objectives

- To recognize bony and soft tissue injuries to the cervical spine
- To appreciate complications related to cervical spine injuries
- To understand management options in spinal injuries
- To recognize bony and soft tissue injuries to the lumbar spine
- To identify urgent and emergent spinal conditions
- To interpret clinical signs and symptoms of spinal pathologies

RED WOLVES and BEARS

**NEW YORK INSTITUTE
OF TECHNOLOGY**

College of Osteopathic
Medicine



Outline

- History and Physical
- Imaging and Labs
- Definitions
- Cervical Orthopedic Disorders
- Thoracic / Lumbar Orthopedic Disorders

History- Spine

- MC chief complaint relating to the spine? Pain
- What are the different kinds of pain? O P Q R S T
 - Localized pain (Tumor/Infection)
 - Mechanical pain (Fracture, Discogenic Disease)
 - Radicular pain (Herniated disc, stenosis)
 - Night pain (tumor)
- What else is important to cover in your history
 - Systemic Symptoms (Fever, Weight Loss, Mental Status Changes) (ROS)
 - Neurologic symptoms (OPQRST, ROS)
 - Trauma history (MIMASH)
 - Occupations (Identify work related injuries, current job and past jobs) (FED TACOS)
 - Recreational activities (FED TACOS)

Physical Exam

- Inspection
- Gait Analysis
- Palpation of the spine
- Range of Motion
- Neurologic Assessment of trunk and extremities
- Shoulder & Hip evaluation depending on the Chief Complaint
 - Cervical CC → Examine shoulders
 - Lumbar CC → Examine hip and lower extremities
- Special Testing

Imaging Studies and Lab work

- Plain Radiographs (X-ray; conventional radiography)
 - Usually 1st image of choice
 - Can identify acute fractures
- CT Scan
 - More details on bone anatomy
 - Pre-op planning, post-op evaluation, trauma fx
- MRI
 - Better soft tissue resolution
 - Herniated discs, degenerative disease, stenosis, tumor, infection, spinal ligament injury

Imaging Studies and Lab work

- Bone Scan (Skeletal Scintigraphy)
 - Nuclear medicine test (uses radioactive tracer)
 - Tracer locates areas of Increased metabolic activity
 - Metastatic Disease, Infection, stress fracture
- Blood Tests
 - CBC, CMP, CRP, ESR, SPEP/UPEP

Cervical Strain / Cervical Sprain

- Etiology
 - Soft tissue activity threshold is surpassed
 - Whiplash Injury (Acceleration-Deceleration Injury)
 - Many cervical muscles do not terminate in tendons but attach directly to the periosteum. Muscles respond to injury by contracting
- Findings
 - Non-focal (distributed throughout), non-radiating neck pain
 - Spasm of paraspinal muscles and trapezius
 - Decreased Range of Motion
 - Upper back and shoulder girdle pain
- Associated with occipital headaches (occipital neuralgia)

Whiplash X-ray

Radiograph of the lateral cervical spine shows a **normal** lordotic curve



Radiograph of the lateral cervical spine shows **straightening** of the lordotic curve



**NEW YORK INSTITUTE
OF TECHNOLOGY**

College of Osteopathic
Medicine

Photos:

<http://emedicine.medscape.com/article/306176-overview>

Cervical Spondylosis

- Etiology
 - Chronic disc degeneration and associated facet arthropathy
 - Wear and tear phenomena that occur across the disc-vertebrae interface
 - Usually associated with aging and the inherent loss of water content within the disc itself, becoming stiffer
- Results in **Discogenic neck pain**
 - Insidious onset neck pain, **without FNS**, exacerbated by vertebral motion

Sources: 1. Stockard, Alan, Osteopathic Clinical Joint Exam; American Osteopathic Academy of Sports Medicine; Chapter 1

2. Miller Essential Orthopaedics, 2020. Chapter 108

Cervical Spondylosis

- Progresses to **radiculopathy**
 - **U/L** neck/shoulder/arm pain, paresthesia, or numbness
 - Exacerbated by mechanical stress on nerve root (motion)
 - **Spurling's Maneuver**
- Severe and chronic cases result in **myelopathy**
 - **Bilateral** weakness (**Upper > Lower**), decreased manual dexterity, **gait disturbances**, sensory defects, **fine motor skills difficulty**, bowel/bladder dysfunction

Cervical Spondylosis

**NEW YORK INSTITUTE
OF TECHNOLOGY**

College of Osteopathic
Medicine



Sources: Miller Essential Orthopaedics, 2020

<https://www.linkedin.com/pulse/20140917125222-154514887-cervical-spondylosis-knowing-what-it-is-a-way-out>

<https://orthoinfo.aaos.org/en/diseases--conditions/cervical-spondylotic-mylopathy-spinal-cord-compression/>

Cervical Disc Disease

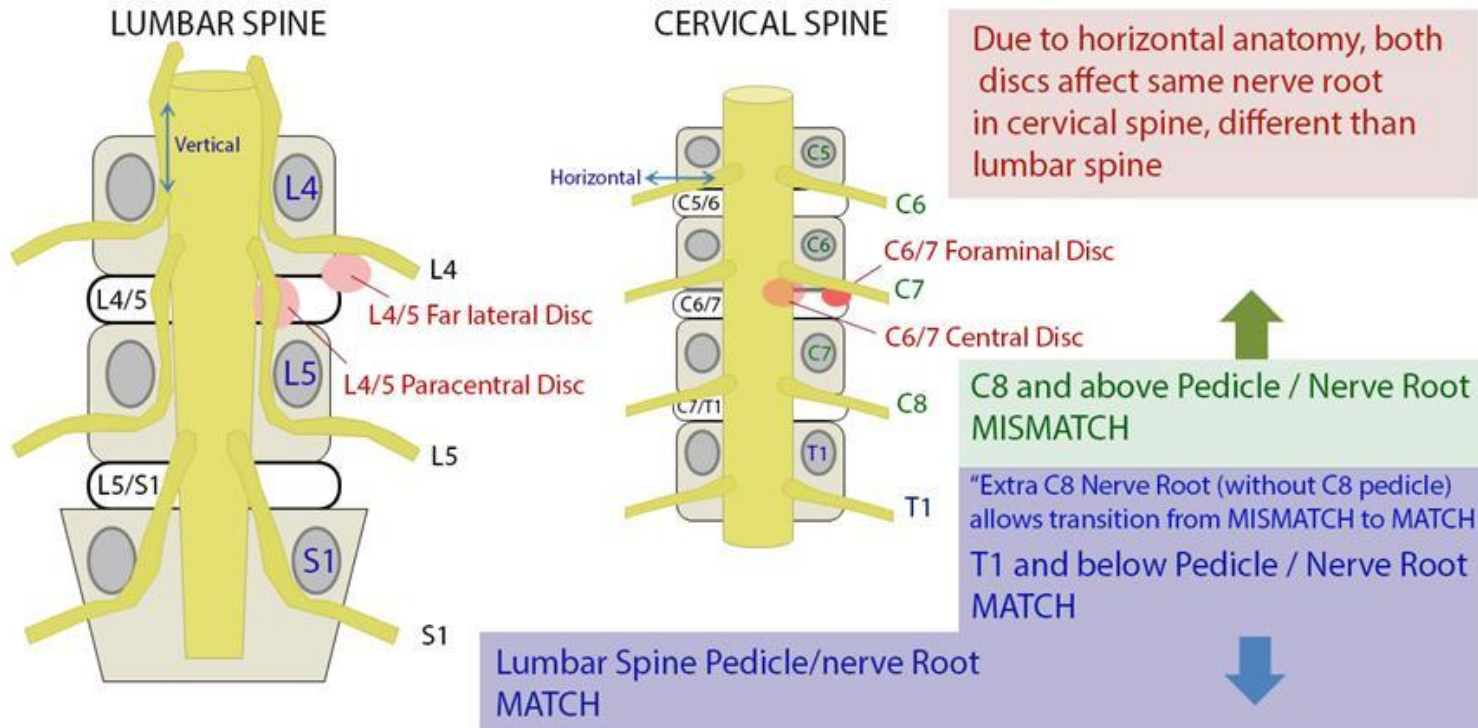
- Etiology
 - Degenerative cervical spondylosis
 - Disc herniation
- Pathophysiology
 - Nerve root irritation via direct compression and inflammatory cytokines
 - Affects nerve root below (C6/7 will affect C7 nerve root)
- Presentation (very similar to cervical spondylosis)

Sources: Stockard, Alan, Osteopathic Clinical Joint Exam; American Osteopathic Academy of Sports Medicine; Chapter 1
Miller Essential Orthopaedics, 2010
<https://www.orthobullets.com/spine/2030/cervical-radiculopathy>

Cervical Disc Disease

NEW YORK INSTITUTE
OF TECHNOLOGY

College of Osteopathic
Medicine

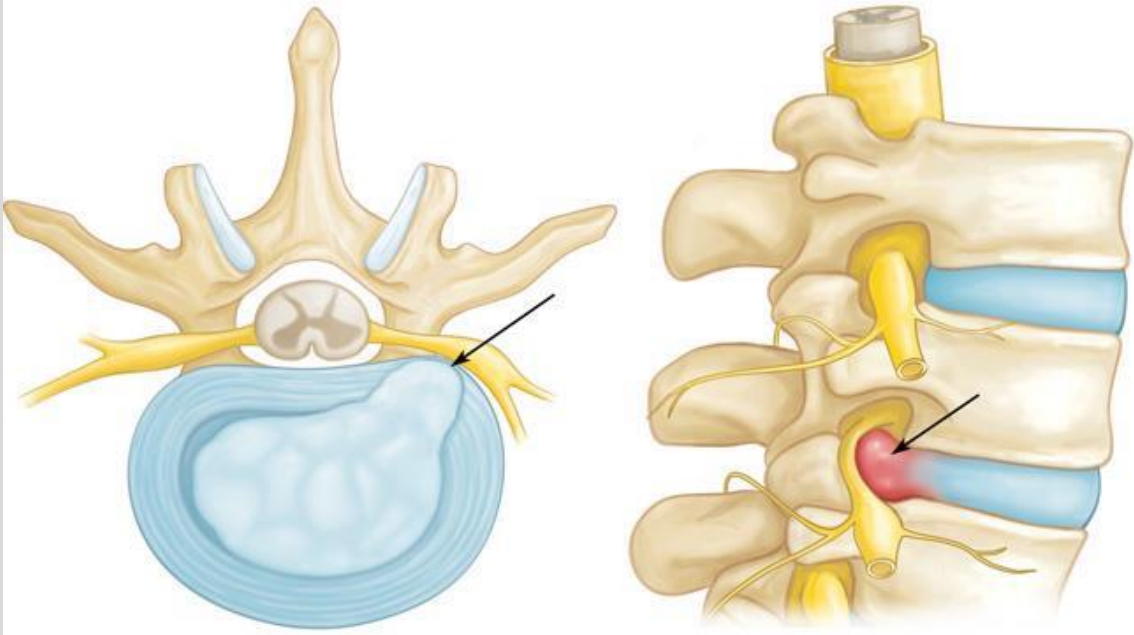


Source:
<https://www.orthobullets.com/spine/2030/cervical-radiculopathy>

Herniated Disc (Herniated Nucleus Pulposus; HNP)

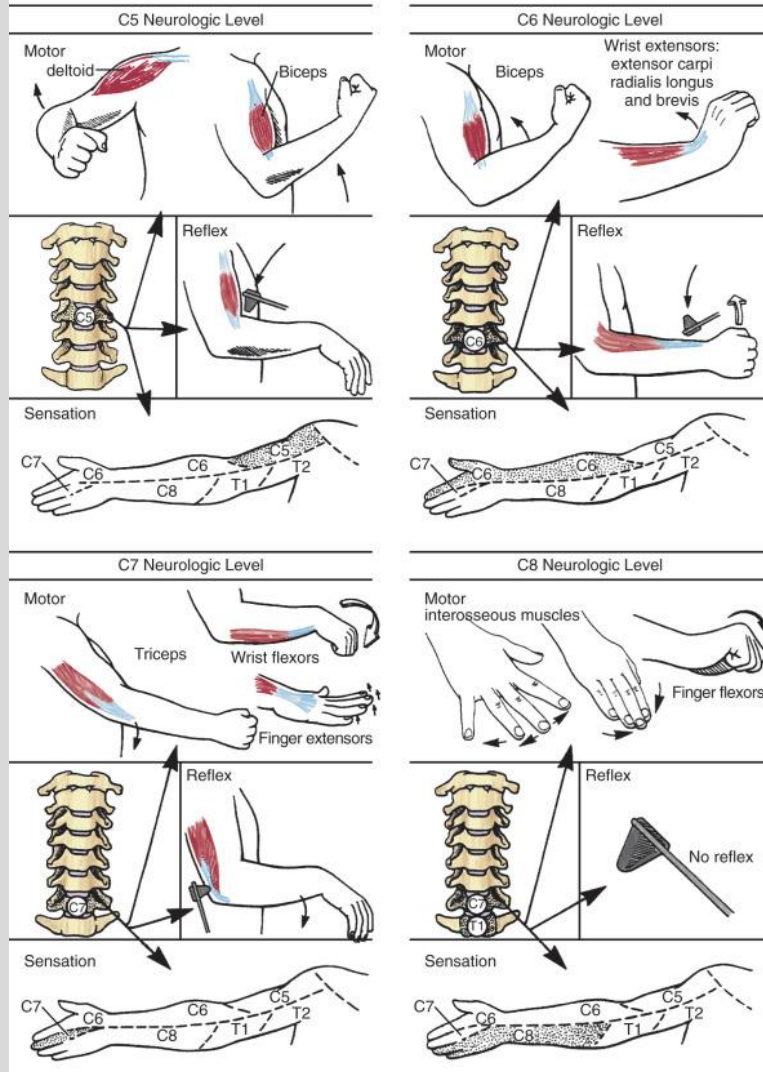
**NEW YORK INSTITUTE
OF TECHNOLOGY**

College of Osteopathic
Medicine



Source: <https://orthoinfo.aaos.org/en/diseases--conditions/cervical-spondylotic-myelopathy-spinal-cord-compression/>

Cervical Disc Disease-Physical Exam



**NEW YORK INSTITUTE
OF TECHNOLOGY**

College of Osteopathic
Medicine

Source: Miller Essential
Orthopaedics, 2020. Chapter
108: "Cervical Disc Disease
and Spondylosis"

Cervical Disc Disease- Physical Exam

- Special Testing for:
 - Radiculopathy: hyporeflexia and atrophy
 - Spurling's sign
 - Shoulder abduction test
 - Myelopathy: hyperreflexia, positive Babinski's sign, clonus
 - Hoffmann's sign
 - Grip and release test
 - Lhermitte's sign

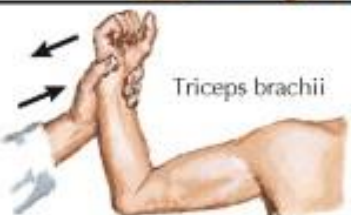
Sources: Miller Essential Orthopaedics, 2020. Chapter 108.

<https://www.orthobullets.com/spine/2030/cervical-radiculopathy>

Cervical Disc Disease- Physical Exam

**NEW YORK INSTITUTE
OF TECHNOLOGY**

College of Osteopathic
Medicine

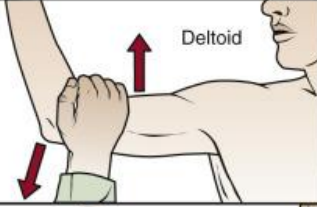
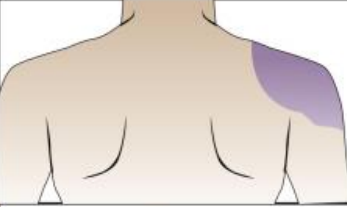
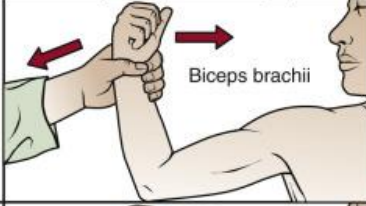
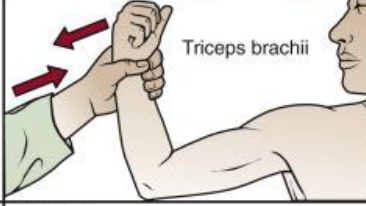
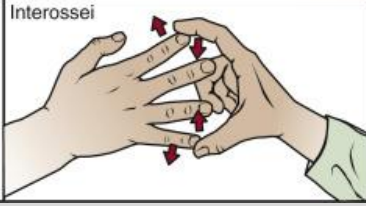
Level	Motor	Reflex	Sensory
C5	 Deltoid	 Biceps brachii	
C6	 Biceps brachii	 Brachioradialis	
C7	 Triceps brachii	 Triceps brachii	
C8	 Interossei	 None	

Source: Thompson, Jon
C. Netter's Concise
Orthopaedic Anatomy,
Chapter 2, 29-74. 2010.

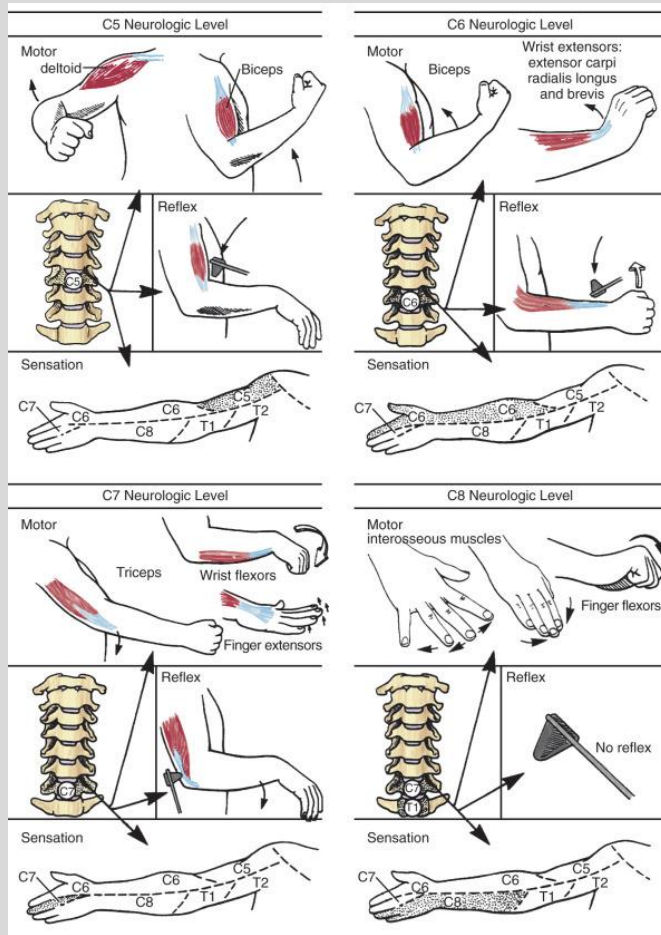
Cervical Disc Disease- Physical Exam

**NEW YORK INSTITUTE
OF TECHNOLOGY**

College of Osteopathic
Medicine

Level	Motor signs (weakness)	Reflex signs	Sensory less
C5	 Deltoid	N/A	
C6	 Biceps brachii	 Biceps brachii Weak or absent reflex	
C7	 Triceps brachii	 Triceps brachii Weak or absent reflex	
C8	 Interossei	N/A	

Source: Miller Essential
Orthopaedics, 2020.
Chapter 106: “Overview
of the Spine”



Source: Miller Essential Orthopaedics, 2020.
Chapter 108: "Cervical Disc Disease and Spondylosis"

NEW YORK INSTITUTE OF TECHNOLOGY

College of Osteopathic
Medicine

Level	Motor signs (weakness)	Reflex signs	Sensory less
C5	Deltoid	N/A	Shoulder and lateral arm
C6	Biceps brachii	Biceps brachii	Wrist and lateral forearm
C7	Triceps brachii	Triceps brachii	Medial forearm and hand
C8	Interossei	N/A	Medial hand and fingers

Source: Miller Essential Orthopaedics, 2020.
Chapter 106: "Overview of the Spine"

Cervical Disc Disease- Imaging and Treatment

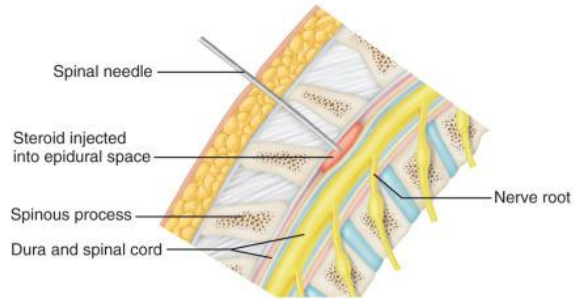
**NEW YORK INSTITUTE
OF TECHNOLOGY**

College of Osteopathic
Medicine

- Radiographs: AP, lateral and oblique views
- CT
 - Shows ossification, foraminal stenosis, and bony cord compress
- MRI
 - Imaging test of choice for cervical, thoracic, and lumbar disc disease
 - Best for spinal cord lesions (syringomyelia, tumors, infection)
- Treatments
 - Conservative
 - NSAIDs (oral steroids if severe), ice/heat, activity modification, soft collar
 - Physical Therapy: Stretching/Traction manipulation/ROM → Isometric exercise / strength → workplace ergonomic mods
 - Surgical
 - Posterior foraminotomy, anterior cervical discectomy & fusion

Source: Miller Essential Orthopaedics, 2020

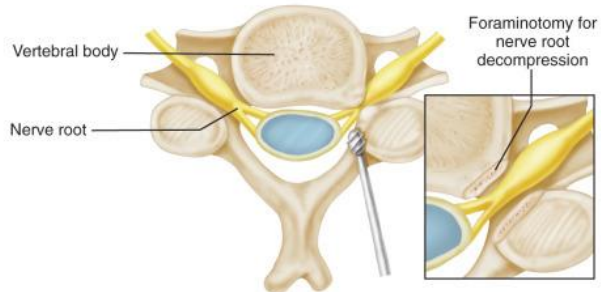
A Epidural Steroid Injection



B Anterior Cervical Discectomy with Fusion or Total Disc Replacement



C Posterior Foraminotomy



NEW YORK INSTITUTE OF TECHNOLOGY

College of Osteopathic
Medicine

Source: Miller Essential
Orthopaedics, 2020. Chapter 108:
“Cervical Disc Disease and
Spondylosis”

<http://arktos.nyit.edu/login?url=https://www-clinicalkey-com.arktos.nyit.edu/#!/browse/book/3-s2.0-C20160046302?indexOverride=GLOBAL>

Rheumatoid Cervical Spondylitis

- ~90% of patients with RA have cervical spine involvement
- Presents with neck pain, stiffness, and headaches
- Neurologic impairment can occur, but it usually gradual
- Most common etiologies
 - Atlantoaxial subluxation: excessive movement between axis & atlas
 - Due to pannus formation between odontoid & C1 → **destructive of transverse ligament of the Atlas**
 - Basilar invagination: invagination where top of C2 migrates upward, narrowing of the foramen magnum

Sources: Moore Clinically Oriented Anatomy, 7th ed

<http://www.svuhradiology.ie/wp-content/uploads/2015/04/RAbasilarimp-1024x562.jpg>

<https://www.orthobullets.com/spine/2043/rheumatoid-cervical-spondylitis?expandLeftMenu=true>

[https://www.uptodate.com/contents/cervical-subluxation-in-rheumatoid-](https://www.uptodate.com/contents/cervical-subluxation-in-rheumatoid-arthritis?search=Rheumatoid%20spine%20disease&source=search_result&selectedTitle=2~150&usage_type=default&display_rank=2#H1)

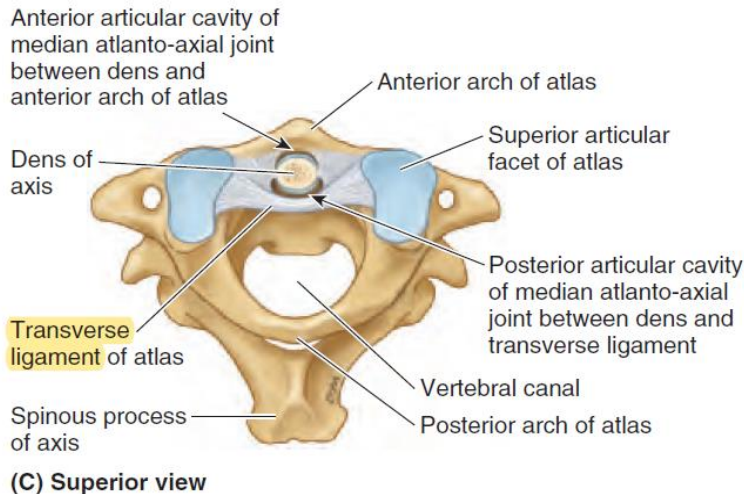
[arthrit](https://www.uptodate.com/contents/cervical-subluxation-in-rheumatoid-arthritis?search=Rheumatoid%20spine%20disease&source=search_result&selectedTitle=2~150&usage_type=default&display_rank=2#H1)

Rheumatoid Cervical Spondylitis

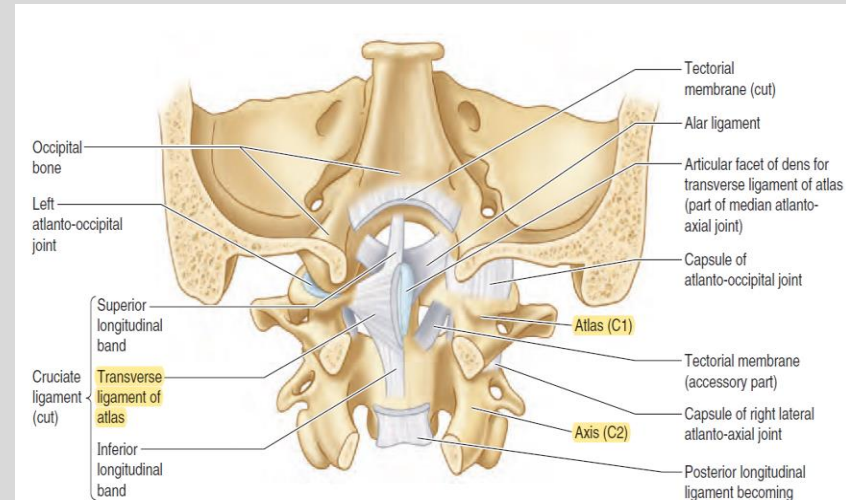
NEW YORK INSTITUTE
OF TECHNOLOGY

College of Osteopathic
Medicine

- In the absence of a competent transverse ligament, the upper cervical region of the **spinal cord** may be **compressed between** the approximated **atlas (C1)** and the **dens (C2)** causing paralysis of all four limbs (quadriplegia), or into the medulla of the brainstem, resulting in death.



Source: Moore
Clinically
Oriented
Anatomy, 7th ed



Summary- Cervical Spinal Disorders

- Cervical Sprain/Strain: Whiplash, loss of cervical lordosis
- Cervical Spondylosis: Chronic DD, Discogenic pain → Radiculopathy → Myelopathy
- Cervical Disc Disease: Spondylosis + Herniation; Nerve Root BELOW
 - C5 – Biceps
 - C6 – Wrist extensors, brachioradialis
 - C7 – Triceps
 - MRI; Start Conservative treatment
- Rheumatoid Cervical Spondylitis
 - C1-C2 subluxation, transverse ligament of Atlas
 - Skull manipulation → quadriplegia

Thoracic / Lumbar Spine

- Vertebral Compression Fracture
- Lumbar Herniated Disc
- Discogenic Back Pain
- Spondylolysis
- Spondylolisthesis
- Ankylosing Spondylitis
- Spinal Stenosis
- Cauda Equina

Vertebral Compression Fracture

- Etiology
 - Fragility fracture of spine in patients with low BMD (Osteoporosis)
- Pathophysiology
 - Decreasing BMD with aging causes decreased bone quantity (mass) not quality
 - Thinned cortical bone; cancellous bone loses trabecular continuity
 - T-Score < 2.5
- Presentation
 - Point tenderness localized to area of fracture (often over SP)
 - Thoracic spine is commonly affected
 - Increased kyphosis (from multiple fractures), loss of height

Vertebral Compression Fracture

- Diagnosis
 - Usually clinical but x-ray of the entire spine needed to assess for concomitant spine fractures
 - CBC, BMP, ESR, Urine and Serum SPEP to r/o infection/malignancy
- Treatment
 - Observation, pain relief, gradual return to activity
 - Bisphosphonates to decrease risk of future fragility fracture
 - Extension orthosis (brace) are beneficial but have poor compliance
 - Surgery: Kyphoplasty (Indicated for severe pain refractory to nonoperative treatment)

Lumbar Herniated Disc (Herniated Nucleus Pulposus; HNP)

NEW YORK INSTITUTE
OF TECHNOLOGY

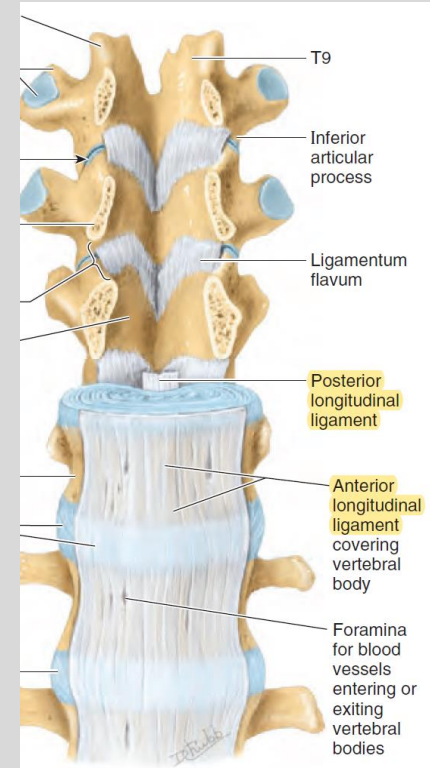
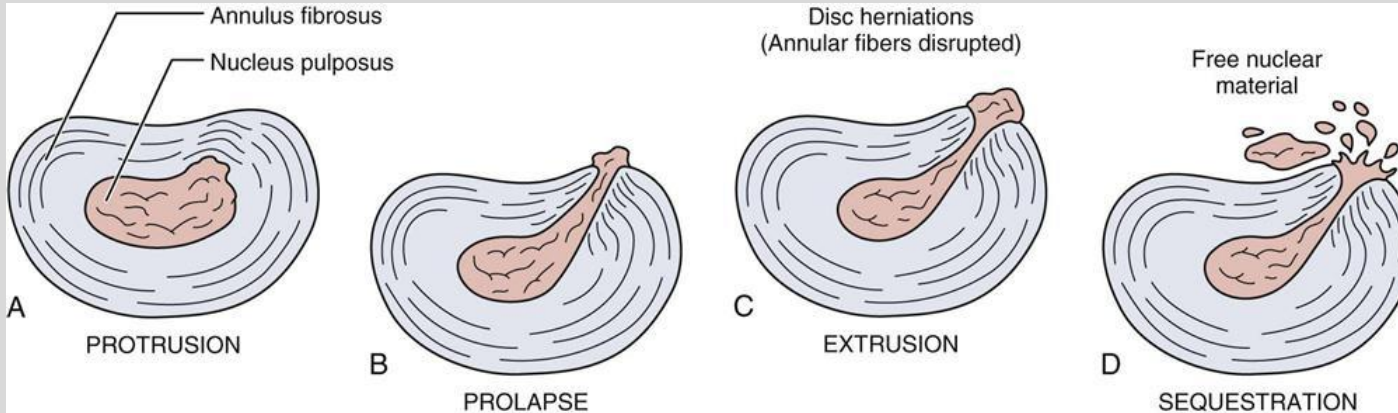
College of Osteopathic
Medicine

- Etiology
 - IV disc degeneration that begins with decreased NP water & proteoglycan content → impairs load-bearing ability
 - Type II collagen (weaker) replaces type I → tears/fissures develop in the AF.
 - Vertebral endplates calcify, impairing nutrient diffusion to the intervertebral disc
 - Progresses to Disc bulging, herniation, facet joint hypertrophy, thickening of ligamentum flavum
- Location
 - Of all symptomatic intervertebral disc herniations, less than 4% occur in the thoracic spine
 - Lumbar herniated NP (HNP); most common at L4-L5, then L5-S1
 - Usually posterolateral (Posterior longitudinal ligament is the weakest)
- If disc impinges nerve roots it causes lumbar radiculopathy
 - I.e. Sciatica if it is one of roots that contributes to the sciatic nerve (L4-S3)

Lumbar Herniated Disc (Herniated Nucleus Pulposus; HNP)

**NEW YORK INSTITUTE
OF TECHNOLOGY**

College of Osteopathic
Medicine



Sources: Miller Essential Orthopaedics, 2020
Moore Clinically Oriented Anatomy, 7th ed

Lumbar Herniated Disc

**NEW YORK INSTITUTE
OF TECHNOLOGY**

College of Osteopathic
Medicine

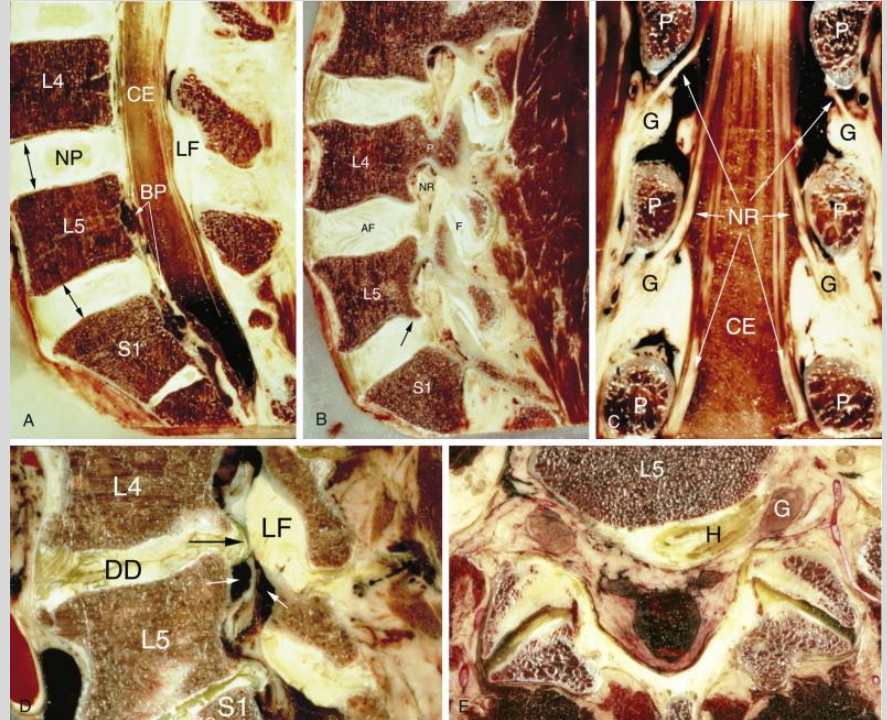
Normal

- NP: Nucleus pulposus
- NR: Nerve Roots
- LF: ligamentum flavum
- CE: cauda equina

Sources: Gallego J, Schnuerer AP, Manuel C: Basic Anatomy and Pathology of the Spine. Medtronic Sofamor Danek, USA, Inc., Memphis, TN, 2001; photographs by Wolfgang Rauschning, MD, PhD. Taken from Miller Essential Orthopaedics, 2020

Degenerative

- DD: degenerative disc
- LF: ligamentum flavum



Lumbar Herniated Disc – Physical Exam

- Observation: changes in symmetry, muscle atrophy, gait analysis
- Palpation
 - Paraspinal muscle spasm/tenderness
 - Sciatic notch tenderness (in radiculopathy)
 - Step-off may indicate associated spondylolisthesis
- ROM testing & special tests

Lumbar Herniated Disc – Physical Exam

**NEW YORK INSTITUTE
OF TECHNOLOGY**

College of Osteopathic
Medicine

Test	Outcome
SLR, sitting and supine	Reproduction/aggravation of radicular pain in exact dermatomal pattern of involved root with elevation of involved leg (sciatic nerve involvement should produce pain distal to knee)
Contralateral SLR	SLR of nonpainful limb causes tension of involved root but from opposite side, causing radiculopathy; may suggest sequestered or extruded large disc fragment
Bowstring sign	Pressure over popliteal fossa aggravates SLR radicular pain
Lasègue's sign	Ankle dorsiflexion aggravates SLR radiculopathy
Femoral nerve stretch test	While patient is prone, examiner extends hip/leg (with or without flexed knee) to stretch femoral nerve roots L2-L4; painful if irritated

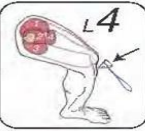
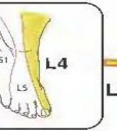
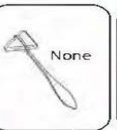
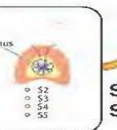
Source: Miller Essential Orthopaedics, 2020

L3 - **L4**

L4 - **L5**

L5 - **S1**

CLINICAL EVALUATION OF NEUROLOGIC LEVELS L4 to S1 (symptoms and signs in extremities)

Mnemonics	Motor	Reflex	Sensation
L4 	 Tibialis anterior	 Patellar tendon	 L4
L5 	 Extensor digitorum longus	 None	 L5
S1 	 Peroneus longus and brevis	 Achilles tendon	 L4 L5 S1
S2-S5 	 Clawing of toes	 Anal "wink"	 S2 S3 S4 S5

achille S1 weak spot

B

Lower Extremity Neurologic Examination

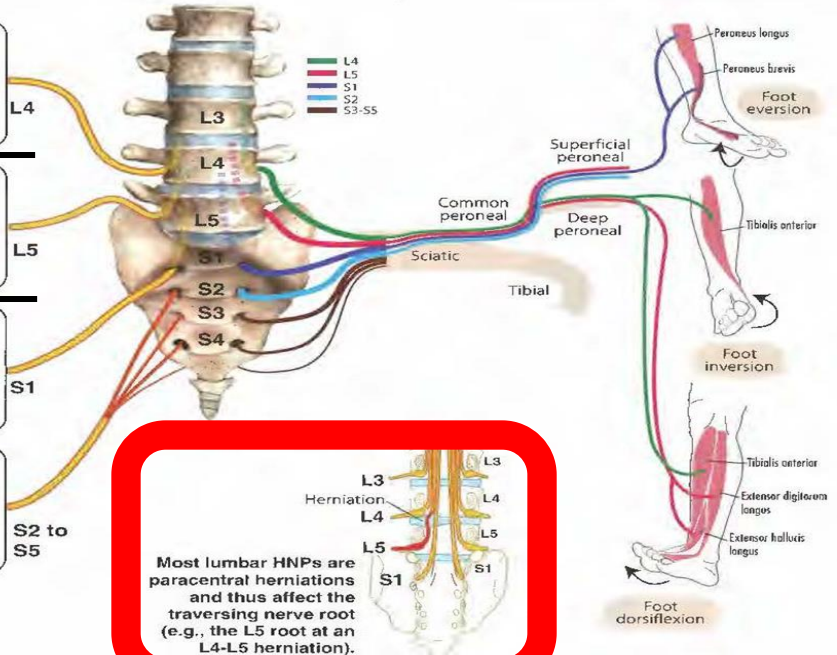


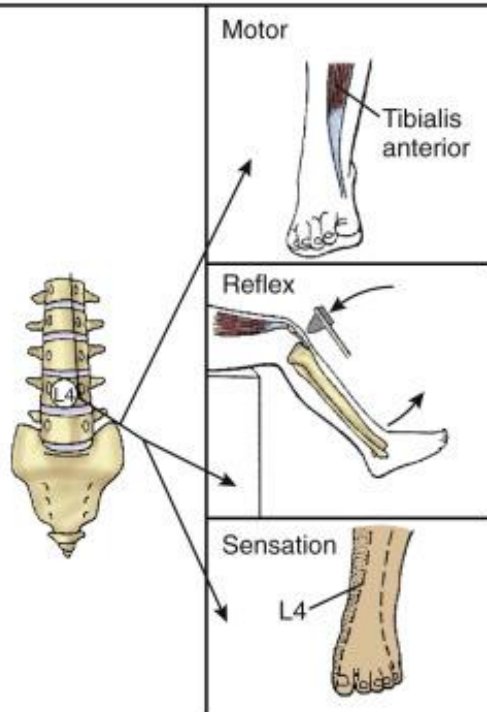
FIGURE 8-1, cont'd

Cervical Disc Disease- Physical Exam

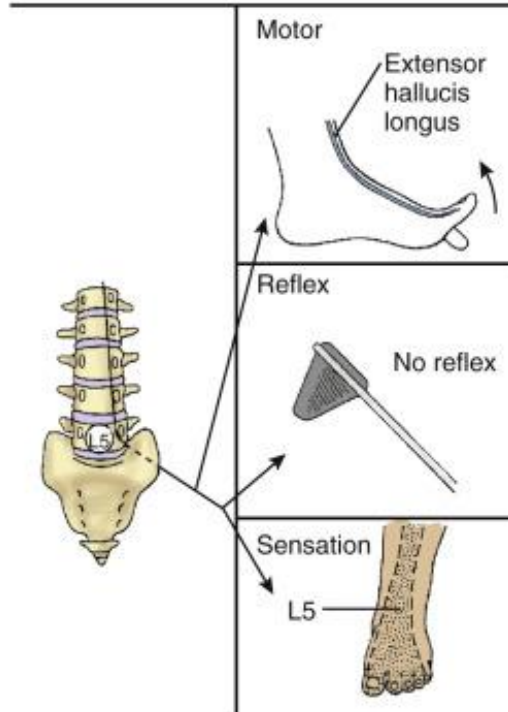
**NEW YORK INSTITUTE
OF TECHNOLOGY**

College of Osteopathic
Medicine

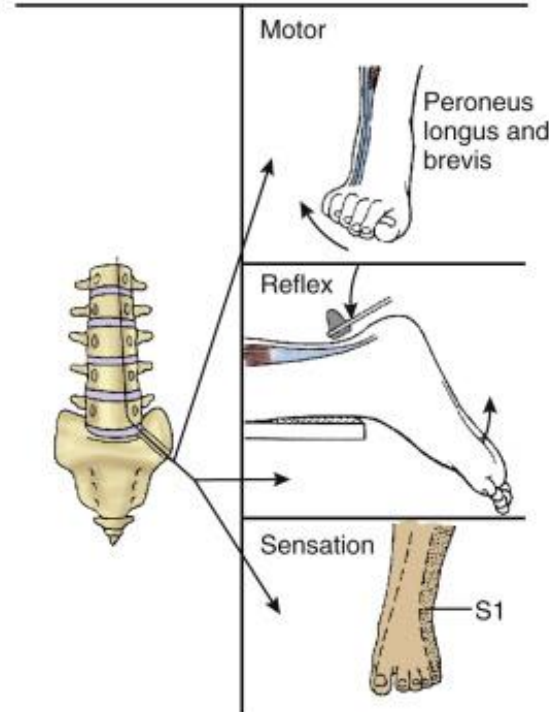
L4 Neurologic Level



L5 Neurologic Level



S1 Neurologic Level

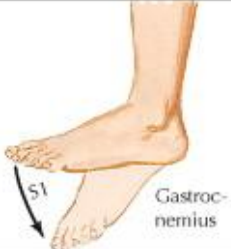


Source: Miller
Essential
Orthopaedics,
2020.
Chapter 113:
"Thoracolumbar
Disc
Disease"

Lumbar Herniated Disc – Physical Exam

**NEW YORK INSTITUTE
OF TECHNOLOGY**

College of Osteopathic
Medicine

Level	Motor	Reflex	Sensory
L4	 Quadriceps Tibialis anterior L4	 Patella tendon ("knee jerk")	 Medial calf/ankle
L5	 Extensor hallucis longus <i>F. Netter M.D.</i>	None	 Dorsal foot and 1st web space
S1	 Gastrocnemius S1	 Achilles tendon ("ankle jerk")	 Plantar and lateral foot

Source: Thompson, Jon C. Netter's Concise Orthopaedic Anatomy, Chapter 2, 29-74. 2010.

Lumbar Herniated Disc – Diagnostic Imaging & Testing

**NEW YORK INSTITUTE
OF TECHNOLOGY**

College of Osteopathic
Medicine

- **Imaging Study of Choice** assessment: **MRI**
 - Better than CT b/c it can also find other intraspinal pathologies (inflammation, malignancy, vascular disease, etc) which can mimic HNP
 - Lots of false positives (ie abnormal findings in asymptomatic patients which are NOT predictive of future back pain)
 - **Non-invasive, no contrast, no radiation**
- **CT myelography**
 - Dural puncture and injection of **contrast** into thecal sac. Outlines spinal cord.
 - Patients with **contraindications to MRI** (e.g. implanted pacemakers or defibrillators)
- **Nerve Conduction Studies (NCS) + Electromyography (EMG)**

Sources: Miller Essential Orthopaedics, 2020

https://openi.nlm.nih.gov/detailedresult.php?img=PMC4300893_IJSS-6-2012-06-001-g007&req=4

https://www.uptodate.com/contents/acute-lumbosacral-radiculopathy-pathophysiology-clinical-features-and-diagnosis?search=diagnosis%20of%20herniated%20lumbar%20disc&source=search_result&selectedTitle=1~29&u sage_type=default&display_rank=1#H23

Lumbar Herniated Disc – Treatment

- Non-operative Treatment (**90% of patients recover in 3 months**)
 - Activity modification, NSAIDS, Ice/Heat
 - PT emphasis on core & lumbar strengthening/stabilization
 - Refractory cases: epidural steroid injections for radiculopathy NOT axial back pain
- Operative Treatment indicated for:
 - Neurologic deficits: spinal cord dysfunction/myelopathy, severe motor deficit, cauda equina syndrome
 - Failed non-operative tx
 - Mitigating psychosocial factors
- Surgical Procedures
 - **HNP** + radiculopathy: **discectomy** (remove disc fragment); >90% success
 - **DDD** with axial low back pain: **fusion**, disc replacement; results inconsistent

Discogenic Back Pain

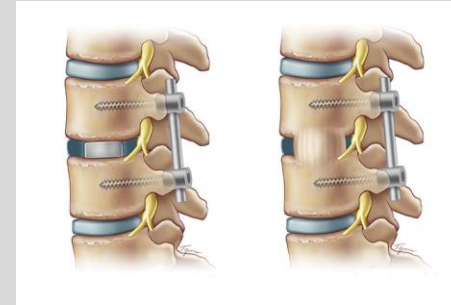
- Etiology
 - Disc degeneration and wear, without herniation
- Presents as:
 - Axial low back pain without radiculopathy
 - Pain exacerbated by
 - Bending
 - Sitting
 - Axial loading
- Examination
 - Nerve tension (ex. SLR) signs are negative
- Diagnostic Imaging
 - Radiographs may show disc space narrowing
 - MRI shows DD without significant stenosis or herniation
- Treatment is usually conservative, but operative options are available



Discogenic Back Pain- Interbody Fusion Surgery

- Spinal Fusion

- m/c surgery for chronic nonspecific LBP with lumbar Disc Disease
- Fuses two or more vertebral bodies together
- Goal: **Restrict spinal motion** AND **remove the degenerated disc**
- Variety of fusion techniques
- Involves the **placement of a bone graft between the vertebrae**
- Alters normal mechanics of the spine and is associated with an increase in long-term degenerative changes in adjacent spine segments
- American Pain Society → Recommend surgery option to patients with persistent (>1 year) disabling non-radicular LBP with discussions of risks and benefits and with interdisciplinary rehabilitation discussed as similarly effective



Sources: https://www.uptodate.com.arktos.nyu.edu/contents/subacute-and-chronic-low-back-pain-surgical-treatment?search=Interbody%20fusion&source=search_result&selectedTitle=1~150&usage_type=default&display=1

<http://weillcornellbrainandspine.org/procedure/tlif-minimally-invasive-spine-fusion>

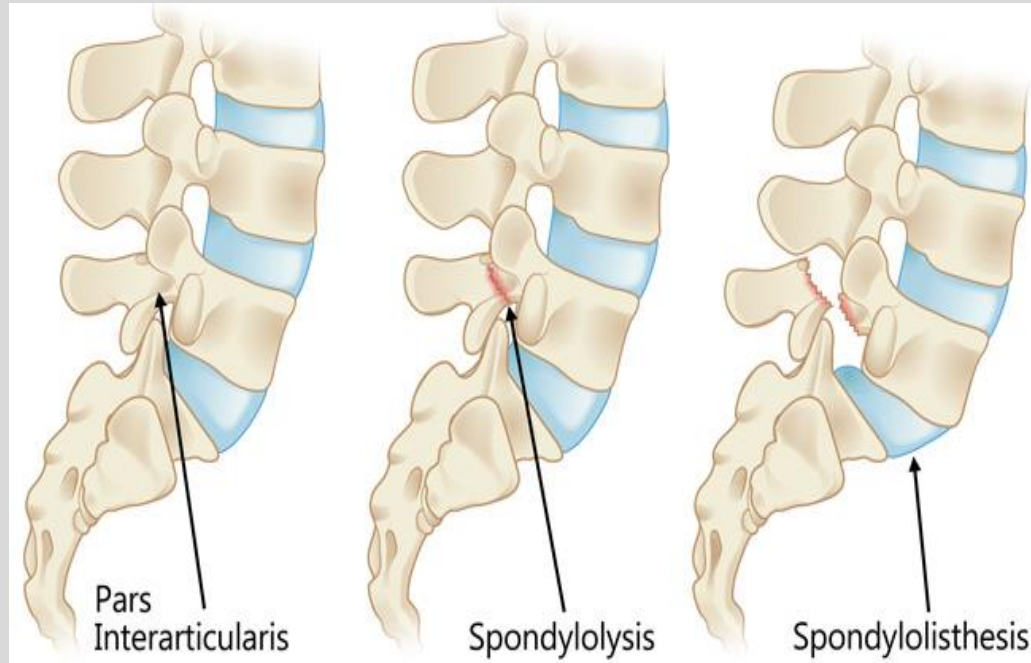
Spondylolysis and Spondylolisthesis

- Spondylolysis: **Stress fracture** in the **pars interarticularis** (junction between superior/inferior facets)
- Spondylolisthesis: **anterior slippage** of one vertebrae on another

Spondylolysis and Spondylolisthesis

**NEW YORK INSTITUTE
OF TECHNOLOGY**

College of Osteopathic
Medicine



Sources: <http://orthoinfo.aaos.org/topic.cfm?topic=a00053>

Spondylolysis

- **Overuse injury** of the **lumbar spine** results from increased loading force on the posterior elements (**pars interarticularis** and the pedicles) with repetitive **extension**
- Usually involved activities where the lower back is repetitively hyperextended
 - Gymnastic, Tennis, Volleyball, Dance, Football (lineman)
 - Specific sports-related maneuvers
- Stress injury → Stress fracture

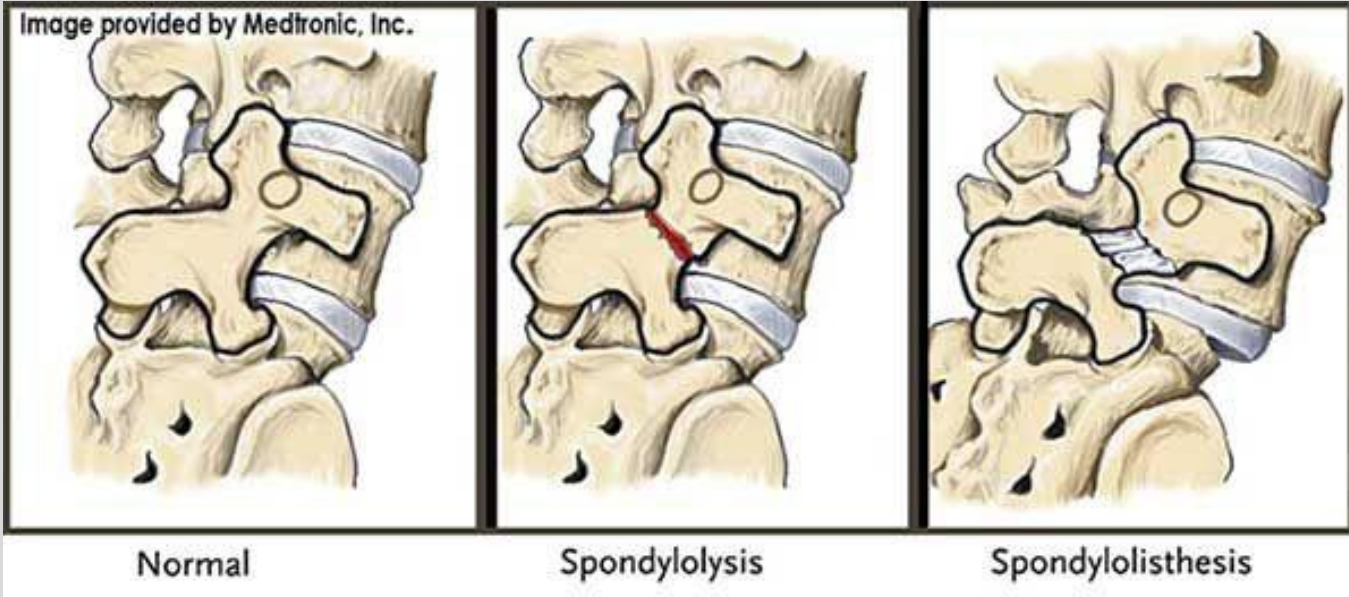
Spondylolysis

- Radiographic workup AP, Lateral, and Oblique views of the lumbosacral spine
- Lateral x-ray
 - Degree of slipping of one vertebra on another
- Oblique view, taken at 45 degrees from the midline
 - Shows the fracture of the pars interarticularis looking like the “collar” on the neck of the “scotty dog”

Spondylolysis

**NEW YORK INSTITUTE
OF TECHNOLOGY**

College of Osteopathic
Medicine



Sources: <https://www.spinemd.com/symptoms-conditions/spondylolysis>

Spondylolysis

Normal



**NEW YORK INSTITUTE
OF TECHNOLOGY**

College of Osteopathic
Medicine

Abnormal



Spondylolysis - Treatment

- Relative Rest
- Stop the specific activity or sport
- NSAIDs / Tylenol
- RTP is based on resolution of pain and degree of spondylolisthesis (if applicable)
- Physical Therapy
- Consideration for a brace
- Gradual return to play
- In severe cases → surgical stabilization

Spondylolisthesis

- One vertebral body is slipping relative to another
- 5 Types
 - Dysplastic (abnormal anatomy)
 - Isthmic (caused by a defect in the pars)
 - Most common in young athletes
 - Degenerative (caused by lax facet joints and a degenerative disc that allows the slippage)
 - More common with age
 - Traumatic (high velocity injuries)
 - Pathologic (tumors or osteopenic bone)

Spondylolisthesis

- Presentation
 - May be asymptomatic
 - May have back pain worsening with extension activities
 - Limited in activities due to back pain, stiffness, and hamstring tightness
- “Step-off” palpable at the deformity
- Grading: % anterior displacement of the fractured vertebrae on the one below
 - 1: < 25%
 - 2: 25-50%
 - 3: 50-75%
 - 4: 75-100%
 - 5: > 100% (Complete displacement)

Question #1

A 25 year-old man comes to the office due to progressive low back pain and stiffness. The pain has been present for several years but has worsened in recent weeks. The patient has pain relief with physical activity and OTC ibuprofen. Associated symptoms include morning stiffness, that lasts an hour. Since last week he has had some eye redness and blurry vision however he has no rash, urinary symptoms or diarrhea. On exam, he is afebrile. There is limited ROM of the lumbar spine with forward flexion as well as tenderness over the lumbosacral area. What is the most likely diagnosis?

- A. Epidural abscess
- B. Rheumatoid arthritis
- C. Reactive arthritis
- D. Ankylosing Spondylitis
- E. Osteoarthritis of the spine

Ankylosing Spondylitis

Autoimmune inflammatory arthritis of the spine

Presentation

- age < 40 (peak incidence between 20-30 y/o)
- Chronic, progressive low back and/or hip pain and stiffness
- Pain improves with activity
- Pain at night

Source: https://www.uptodate-com.arktos.nyit.edu/contents/clinical-manifestations-of-axial-spondyloarthritis-ankylosing-spondylitis-and-nonradiographic-axial-spondyloarthritis-in-adults?search=ankylosing%20spondylitis&source=search_result&selectedTitle=1~150&usage_type=default&display_rank=1#H1

<https://emedicine.medscape.com/article/1209595-clinical>

Ankylosing Spondylitis

Autoimmune inflammatory arthritis of the spine

Exam findings

- Arthritis (**sacroiliitis**) & lumbosacral tenderness
- **Reduced chest expansion & spinal ROM** → causes **pulmonary compromise**
- Enthesitis (tenderness at tendon insertion sites)
- Dactylitis (swelling of fingers & toes)
- Osteopenia/porosis → vertebral fragility fractures → spinal cord and/or nerve root compression,
- Non-articular disease:
 - systemic symptoms (fever, chills, fatigue, weight loss)
 - **anterior uveitis**, psoriasis, IBD
 - aortitis → aortic root dilation → **Aortic regurgitation**

Source: https://www.uptodate-com.arktos.nyit.edu/contents/clinical-manifestations-of-axial-spondyloarthritis-ankylosing-spondylitis-and-nonradiographic-axial-spondyloarthritis-in-adults?search=ankylosing%20spondylitis&source=search_result&selectedTitle=1~150&usage_type=default&display_rank=1#H1

<https://emedicine.medscape.com/article/1209595-clinical>

Ankylosing Spondylitis

Autoimmune inflammatory arthritis of the spine

Labs

- Elevated acute phase reactants (**ESR, CRP**)
- This one of the **SERONEGATIVE** spondyloarthropathies → NO AUTOANTIBODIES will be in present in serum
- **+ HLA-B27** (>90% of patients with AS have HLA-B27)

Source: https://www.uptodate-com.arktos.nyit.edu/contents/clinical-manifestations-of-axial-spondyloarthritis-ankylosing-spondylitis-and-nonradiographic-axial-spondyloarthritis-in-adults?search=ankylosing%20spondylitis&source=search_result&selectedTitle=1~150&usage_type=default&display_rank=1#H1

<https://emedicine.medscape.com/article/1209595-clinical>

Ankylosing Spondylitis- Bamboo Spine

**NEW YORK INSTITUTE
OF TECHNOLOGY**

College of Osteopathic
Medicine



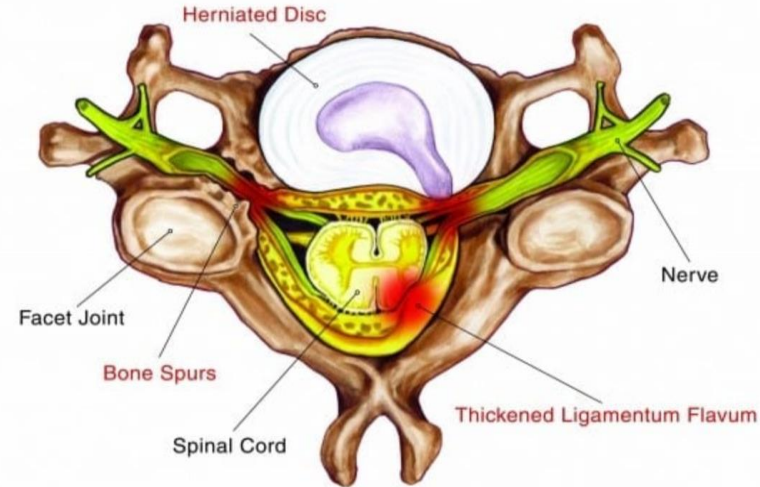
Source: doctorakerkar.wordpress.com/2013/09/28/how-does-ankylosing-spondylitis-progress/

Question #2

- 67 y/o woman comes to the office for evaluation of low back pain and difficulty walking for 6 months. The pain is worse with prolonged standing and is relieved by sitting. The pain is less severe when walking uphill than walking downhill. On examination, there is no spinal tenderness and a negative SLR. Reflexes are 2+ in both legs and distal pulses are intact. What is the most likely diagnosis?
 - A. Herniated Lumbar Disc with Radiculopathy
 - B. Cauda Equina Syndrome
 - C. Peripheral Vascular Disease
 - D. Spinal stenosis
 - E. Vertebral Compression Fracture

Spinal Stenosis

- Etiology
 - **Narrowing** of the **spinal canal** or **neuroforamen** → compression of the thecal sac in central stenosis and/or spinal nerve roots in lateral stenosis.
 - Mostly due to degenerative disease (age related)
 - IV disc herniation &/or shrinkage
 - **Facet Joint arthritis** → bone spurs
 - **Ligamentum flavum hypertrophy**



Spinal Stenosis

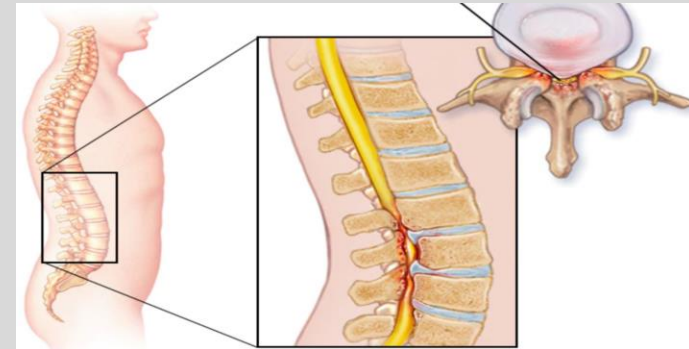
- Classic presentation
 - Elderly Patient
 - Low back or buttock pain +/- LE radicular pain
 - **Neurogenic claudication**: lower extremity pain, paresthesia, and/or weakness with walking/standing*****can be confused with PVD**
 - Weakness is usually mild
 - DTRs may be diminished
 - Pain **persists @ rest** especially when standing
 - ****distinguishes from vascular claudication**
- Relieved by sitting or leaning forward (**shopping cart sign**)
 - Extension (e.g. standing straight) worsens narrowing → exacerbates symptoms
 - **Flexion** increases space in canal/foramina → **decreases compression** of spinal cord/nerve roots
 - ****distinguishes from vascular claudication**

Spinal Stenosis

**NEW YORK INSTITUTE
OF TECHNOLOGY**

College of Osteopathic
Medicine

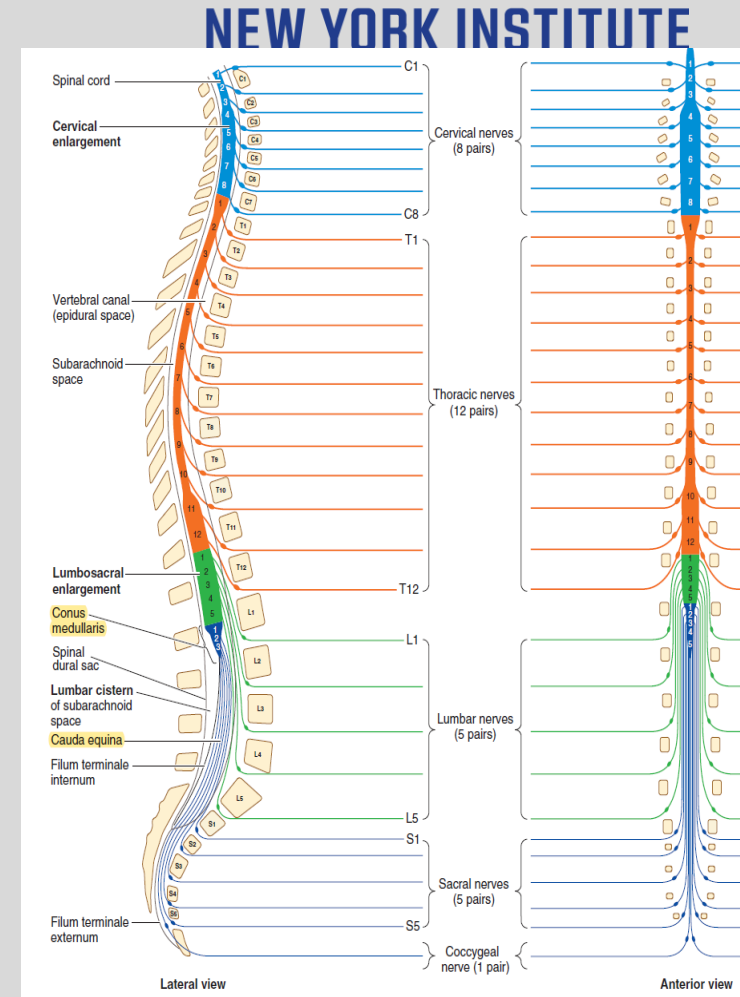
- Imaging studies (XR, CT/MRI)
 - Can be used to confirm stenosis with the proper clinical findings & plan operative treatment if indicated



Cauda Equina Syndrome

- Spinal cord ends at the **conus medullaris (L1)**
- Below this level spinal canal filled with L2-S4 nerve roots
 - Free floating nerve roots look like a horse's tail & thus called cauda equina
- Compression leads to **flaccid paralysis** (different from spastic paralysis seen with myelopathy)
 - The cauda equina are **PERIPHERAL** nerves
- Etiologies
 - Massive central lumbar disc herniation
 - Fracture, Trauma
 - Epidural abscess or hematoma
 - Neoplasm
 - Infection

Sources: Miller Essential Orthopaedics, 2020. Chapter 114.
Moore Clinically Oriented Anatomy, 7th ed



Cauda Equina Syndrome

- Presentation
 - Pain lower back/ buttocks
 - Usually unilateral with severe radicular pain
 - Saddle anesthesia
 - Loss of anal sphincter tone & wink reflex
 - Bowel incontinence
 - Urinary retention → Overflow urinary incontinence
 - Flaccid paralysis & decreased LE reflexes
- Diagnostic Imaging
 - MRI (test of choice)
 - CT Myelography (if unable to do MRI)
- Surgical Emergency!!!
 - Decompression of cauda equina within 48hrs of symptoms onset

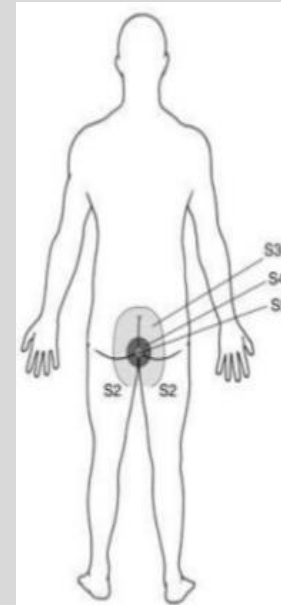


Illustration of saddle anesthesia;
- The S5, S4, and S3 nerves provide sensory innervation to the rectum, perineum, and inner thigh.

Source:
Journal of the American Academy of Orthopaedics Surgeons,
<http://www.jaaos.org>

Cauda Equina Syndrome- MRI



**NEW YORK INSTITUTE
OF TECHNOLOGY**

College of Osteopathic
Medicine

Source:

https://images.radiopaedia.org/images/940993/f7a8adca63efae788f621869cc21e8_big_gallery.jpg

Summary- Thoracic/Lumbar Spinal Disorders

- **Vertebral Compression Fracture:**
 - Osteoporosis, SP point tenderness, thoracic spine, bisphosphonates
- **Lumbar Herniated Disc: NP/AF degen → posterolateral herniation**
 - L4 – Quad weakness, patellar reflex
 - L5 – dorsiflexion, heel walking
 - S1 – plantarflexion, toe walking, “achilleS-1 weakness/reflex”
- **Discogenic Back Pain:**
 - DDD w/o herniation or radiculopathy
- **Spondylolysis:**
 - hyperextension stress fx, “scotty dog”-sign, gymnast
- **Spondylolisthesis:**
 - ant. Displacement, “step-off” sign, L4 on L5
- **Ankylosing Spondylitis:**
 - 20-30s, HLA-B27, chest expansion/pulm compromise, uveitis, aortic regurg, sacroiliitis, bamboo spine
- **Spinal Stenosis:**
 - ANY degenerative narrowing, pain w/ standing, “shopping cart”-sign
- **Cauda Equina**
 - Massive peripheral nerve damage past L1, flaccid, bladder, bowel, saddle anesthesia
 - EMERGENT SURGICAL DECOMPRESSION!!!

Lecture Summary

- Be familiar with the history, physical, and basic imaging needed for evaluation of the spine
- Cervical Spine
 - Understand the mechanism and imaging of cervical sprain/strain
 - Understand the mechanism and presentation of cervical spondylosis & rheumatoid Spondylitis
 - Know the mechanism, pertinent exam findings, and management of cervical disc disease
- Thoracic/Lumbar Spine
 - Know the mechanism, pertinent exam findings, and management of vertebral compression fractures and herniated disc
 - Recognize discogenic back pain
 - Be able to differentiate between spondylolysis and spondylolisthesis
 - Understand the presentation, exam findings, labs and imaging of ankylosing spondylitis
 - Understand the mechanism and presentation of spinal stenosis
 - Recognize the presentation of cauda equina syndrome and know the management

Question #3

A 70 y/o female is brought into the ED by EMS for difficulty breathing. She has been battling a respiratory viral infection for the last few days that has since worsened. On physical exam, you notice B/L swelling of the MCP and PIP joints of the hand. Her daughter is present and tells you that her mother has a hx of atrial fibrillation, stroke, and autoimmune disease. Shortly after arriving to the hospital, the patient begins to decompensate and you worry that she will soon be unable to breathe for herself. You call the rapid response team and proceed with intubation. The following day, while in the ICU, the patient regains consciousness however she is now unable to move her arms or legs. What is the most likely cause of the patient's quadriplegia?

- A. Ruptured berry aneurysm
- B. Embolic stroke
- C. Spinal Cord compression between C1 and C2
- D. Cervical disc herniation
- E. Infectious CNS Abscess

Question #4

Walter White is a 52 year old high school chemistry teacher who moonlights in fast food restaurants for extra money. He comes into your office c/o right arm pain for last few months. He states that he has been having trouble lifting heavy packages and using the cash register with that arm. On physical exam you notice weakness in right wrist extension as well as sensory defects over the lateral right forearm. You are able to reproduce Walter's pain by extending and rotating his neck followed by the addition of a compressive force on his head.

What would you diagnose Walter with?

- A. C4/C5 disc disease with radiculopathy
- B. C5/C6 disc disease with radiculopathy
- C. Cervical spondylosis with myelopathy
- D. C6/C7 disc disease with radiculopathy

Question #5

Walter White is a 52 year old high school chemistry teacher who moonlights in fast food restaurants for extra money. He comes into your office c/o right arm pain for last few months. He states that he has been having trouble lifting heavy packages and using the cash register with that arm. On physical exam you notice weakness in right wrist extension as well as sensory defects over the lateral right forearm. You are able to reproduce Walter's pain by extending and rotating his neck followed by the addition of a compressive force on his head.

What would you expect to find on reflex testing?

- A. Bicep hyporeflexia
- B. Bicep hyperreflexia
- C. Brachioradialis hyporeflexia
- D. Brachioradialis hyperreflexia
- E. Tricep hyporeflexia
- F. Tricep hyperreflexia

References

- Miller, Mark D., Essential Orthopaedics. 2nd Edition. Elsevier, Inc. 2020.
- <https://login.nyit.idm.oclc.org/login?url=https://www-clinicalkey-com.nyit.idm.oclc.org/#!/browse/book/3-s2.0-C20160046302>
- DeLee, Drez, & Miller's Orthopaedic Sports Medicine, Fifth Edition. Elsevier, Inc. 2020.
- Link:
- <https://nyit.idm.oclc.org/login?url=https://www-clinicalkey-com.nyit.idm.oclc.org/#!/browse/book/3-s2.0-C20151044313>
- Madden, Christopher C. Netter's Sports Medicine, Third Edition. Elsevier, Inc. 2023.
- Link:
- <https://nyit.idm.oclc.org/login?url=https://www-clinicalkey-com.nyit.idm.oclc.org/#!/browse/book/3-s2.0-C20190051114>
- Freeman, Michael. "Cervical Sprain and Strain. Link: <http://emedicine.medscape.com/article/306176-overview>
- Stockard, Alan, Osteopathic Clinical Joint Exam; American Osteopathic Academy of Sports Medicine; Chapters 1 and 6
- Metzl, Jordan D., "Sports Medicine in the Pediatric Office." American Academy of Pediatrics.

Lecture Feedback Form:

**NEW YORK INSTITUTE
OF TECHNOLOGY**

College of Osteopathic
Medicine

<https://comresearchdata.nyit.edu/redcap/surveys/?s=HCY448FWYXREL4R>

Thank you for Watching

**NEW YORK INSTITUTE
OF TECHNOLOGY**

College of Osteopathic
Medicine

